

Taking care of supply

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A recurring theme for modern business is adaptability. The ability to exploit resources, skills and market position in new ways to increase returns on investment is paramount. Large companies in particular need to understand their internal and external supply chains to identify and evaluate new opportunities. BT has a history of developing and supplying all the services required by its end customers; but the world is changing rapidly, new types of player are emerging and BT now supplies its many services to many different types of customer — service providers, resellers and peers as well as end customers. There needs to be a way of understanding the implications of the different forms of customer/supplier relationship on the design and implementation of BT's systems.

This paper presents a new set of patterns for businesses and their supply relationship. These patterns can be used to examine supply chains and the paper illustrates how these different supply relationships place different requirements on the operational support systems. Such patterns will be core to a flexible strategy within high performing, adaptable businesses.

1. Introduction

BT's customers are many and varied. They range from residential homes, single person and single premises businesses to those more widely distributed, global businesses, and from businesses that require highly mobile features to those that have fixed requirements. BT co-operates with other companies in the supply of integrated solutions and also aggregates and adds value to others' products to create bespoke solutions. BT provides network connection services and value-added services. How should a company like BT support its many businesses? What sort of support infrastructure should it build? What are the implications of supplying the end customer rather than supplying a reseller and vice versa? This paper offers a way to illuminate the answers to these questions.

A business can be characterised in terms of what it does in order to create value and the relationship a business has with its customers — the responsibilities it undertakes [1, 2]. The creation of value by a business is best modelled as the responsibilities it undertakes on behalf of its customers (e.g. to ensure that a telephone call can be made) and in doing so will require other responsibilities to maintain that relationship (e.g. that the customer is correctly invoiced). The use of stereotypical patterns is proposed; these are termed transformation stamps (describing how value is added) and supply stamps (describing the relationship between supplier and customer). These two elements can then be combined to classify a value-adding enterprise. This combination is referred to as a business pattern. The transformation and supply stamps define ways that

businesses fulfil their resourcing and delivery responsibilities respectively [1].

These business patterns describe what an enterprise does at a very high level and they imply responsibilities that an enterprise must fulfil. The patterns do not attempt to describe processes and systems, they represent an implementation-independent view of an enterprise and are not generated for a specific purpose, e.g. software design [4]. This places them at a higher level of abstraction than other business patterns [5]. Section 2 describes the transformation stamps; section 3 describes the supply stamps and the nature of the relationship between supplier and consumer that characterise these stamps [3]. This paper then extends these ideas as follows:

- the business patterns, and more specifically the supply stamps, can be used to describe supply chains — section 4 gives an example of this and suggests that the supply stamp, via which an enterprise procures items, constrains the supply stamps that they in turn can use to supply their customers,
- although the patterns are implementation independent, the patterns can be used to place processes and systems into a business context — section 5 provides an example of this where the influence that supply stamps have on the requirements of systems that support the commercial interaction is discussed.

2. Transformation stamps

The ways a business can add value to the things it procures can be described in terms of a small number of transformation stamps. Three transformation stamps have been identified — manufacturer, service creator and reseller. A business can adopt more than one of these roles, e.g. BT is primarily a service creator but resells equipment via its retail outlets.

2.1 *Manufacturer*

A manufacturer adds value by changing the nature of raw materials or components it procures to produce a new product. A furniture maker buys wood (raw materials) and sells furniture. In order to achieve this transformation the enterprise procures the equipment and skills required. The ownership of the raw materials transfers to the customer along with the furniture; the machinery remains the property of the manufacturer. The furniture maker makes furniture for many customers and therefore can make furniture more efficiently than if we were all to attempt to make our own furniture (theory of the firm). In a sense we are sharing the skills of the furniture maker and the means of production they own.

2.2 *Service creator*

A service creator procures resources that can be used to provide a service. The service creator then sells rights of access to that resource, thereby creating a service. When the service creator sells its service, there is no transfer of ownership, the resources that provide the service all remain the property of the service creator. The creator of a software-based service, such as Web-page hosting, will procure development tools and machines and the services of programmers. The resulting software will then be deployed on operational machines and form part of the service resource. A Web-page hosting service allows many users to share a server, thus reducing the cost per user. It is this sharing that creates the economies required to create a profitable service. A characteristic of services is that they tend to have large fixed costs; individual customers cannot bear this fixed cost, and therefore it has to be shared among many customers. Service creators enable this sharing. The resource that the service creator procures need not be physical, but may be expertise or information — this is the case for agents and brokers.

2.3 *Reseller*

A reseller buys goods and services in certain quantities at a certain price and sells those goods and services to its customer in different quantities (usually smaller) and at a different price (usually higher). One of the main roles of the reseller is providing access to markets for its suppliers and access to suppliers for its customers.

By supplying to its end customers via resellers, a supplier need only deal with a limited number of other

organisations and a relatively small number of accounts and transactions.

The transformation performed by a reseller is not in the physical nature of the goods but more in their physical location, the quantities in which they are sold, and the other products that are also available from the same location. Resellers allow the end users to procure goods manufactured by a number of different manufacturers in a single transaction.

As the name implies, resellers own what they sell; therefore there is transfer of ownership to the reseller when they purchase stock, then transfer of ownership of that stock when it is sold to the end customer. In fact, resellers do not necessarily sell to end customers — they can sell on to other resellers, as in a wholesaler that supplies to a retailer. The distinction between these two types is in the relationship they have with their customers — this is discussed in section 3.

An important role of a reseller is the acceptance of risk. A reseller will hold stocks that otherwise the manufacturer has to hold, this releases working capital for the manufacturer. There is also a risk that goods/services will not be sold within their shelf life. The retailer bears any loss in this case, again reducing the risk to the manufacturer.

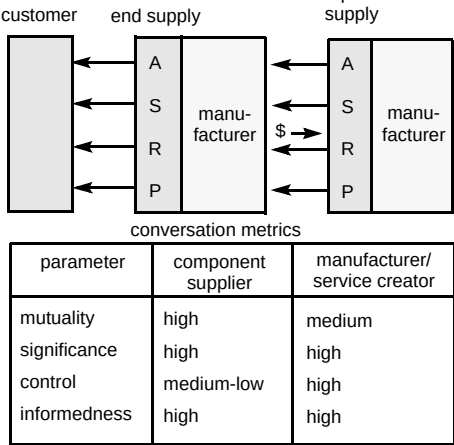
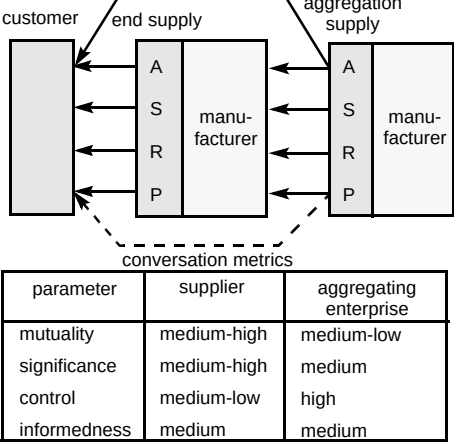
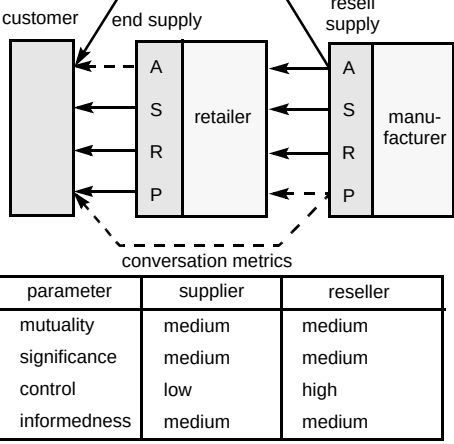
3. Supply stamps

Some goods are supplied to the end customer direct from the manufacturer — however, this is the exception rather than the rule, with many others being supplied via a number of intermediaries. Suppliers take decisions, based on the nature of what is being supplied and the characteristics of the market, about how to supply their product or service and to whom. The supply stamps characterise the various ways that one enterprise can supply another, and therefore how one enterprise can procure from another.

The metrics of mutuality, significance, informedness and control can be used to characterise these customer/supplier conversations [1, 2]. The specifics of the market and the relative power of the two enterprises will determine the values of these metrics, e.g. the control a well-known brand supplier has in supplying a large supermarket is much greater than that of a small local producer supplying that same supermarket; therefore the values shown for these metrics is illustrative rather than definitive.

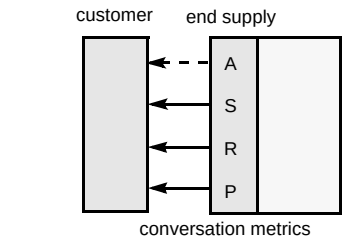
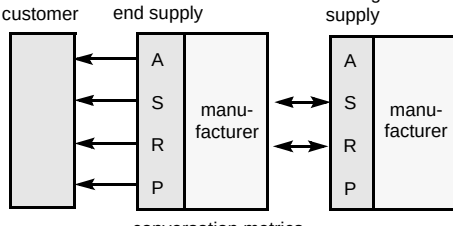
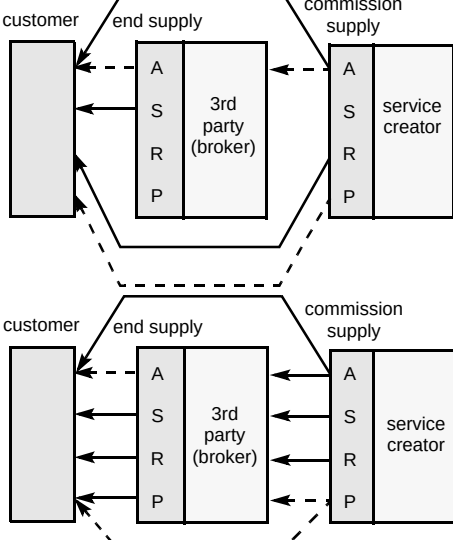
The output stamps are defined by a topology of interaction and the character of the conversation between supplier and consumer. The output stamp(s) to which an enterprise conforms results in a number of implications. Some of these implications are related to the responsibilities that the enterprise has adopted. In the examples in Table 1, the effect each stamp has on the responsibilities for:

Table 1 Supply stamps.

Stamp	Pattern	Examples and notes															
Component supply	 customer end supply component supply conversation metrics <table border="1"> <thead> <tr> <th>parameter</th><th>component supplier</th><th>manufacturer/ service creator</th></tr> </thead> <tbody> <tr> <td>mutuality</td><td>high</td><td>medium</td></tr> <tr> <td>significance</td><td>high</td><td>high</td></tr> <tr> <td>control</td><td>medium-low</td><td>high</td></tr> <tr> <td>informedness</td><td>high</td><td>high</td></tr> </tbody> </table>	parameter	component supplier	manufacturer/ service creator	mutuality	high	medium	significance	high	high	control	medium-low	high	informedness	high	high	The consumed component will become an integral part of the product/service supplied to the end customer. As a failure of the component may cause a failure of the whole product, the component consumer places heavy reliance on the component supplier. Component supply relationships are often long term; this reflects the informedness and significance of this relationship to both parties. As the end customer has no visibility of the component, the customer has no direct interaction with the component supplier. Example: supply of PCBs to a PC manufacturer.
parameter	component supplier	manufacturer/ service creator															
mutuality	high	medium															
significance	high	high															
control	medium-low	high															
informedness	high	high															
Aggregator supply	 customer end supply aggregation supply conversation metrics <table border="1"> <thead> <tr> <th>parameter</th><th>supplier</th><th>aggregating enterprise</th></tr> </thead> <tbody> <tr> <td>mutuality</td><td>medium-high</td><td>medium-low</td></tr> <tr> <td>significance</td><td>medium-high</td><td>medium</td></tr> <tr> <td>control</td><td>medium-low</td><td>high</td></tr> <tr> <td>informedness</td><td>medium</td><td>medium</td></tr> </tbody> </table>	parameter	supplier	aggregating enterprise	mutuality	medium-high	medium-low	significance	medium-high	medium	control	medium-low	high	informedness	medium	medium	Similar to component supply — however, the customer is aware of the element that has been aggregated. This implies that the element supplier needs some direct contact with the end customer, e.g. for some elements of post-sales support. The customer may also have the option to unbundle the aggregated element, perhaps replace one element with an alternative. This is indicative of the looser coupling between aggregated elements of a product compared to components that are an inherent part of an end product. It also highlights the need for enterprises supplying elements to be aggregated to advertise to the end customer. Example: Supply of monitors to PC manufacturers.
parameter	supplier	aggregating enterprise															
mutuality	medium-high	medium-low															
significance	medium-high	medium															
control	medium-low	high															
informedness	medium	medium															
Resell supply	 customer end supply resell supply conversation metrics <table border="1"> <thead> <tr> <th>parameter</th><th>supplier</th><th>reseller</th></tr> </thead> <tbody> <tr> <td>mutuality</td><td>medium</td><td>medium</td></tr> <tr> <td>significance</td><td>medium</td><td>medium</td></tr> <tr> <td>control</td><td>low</td><td>high</td></tr> <tr> <td>informedness</td><td>medium</td><td>medium</td></tr> </tbody> </table>	parameter	supplier	reseller	mutuality	medium	medium	significance	medium	medium	control	low	high	informedness	medium	medium	Products supplied for resell will be sold to the end customer unchanged. As resellers will advertise themselves rather than the products they sell, the manufacturer will need to advertise to the end customer. Part of the reseller's roles is that of pre- and post-sales support, although some aspects of post-sales support will be undertaken by the manufacturer direct (e.g. consumer advice lines). Example: PC manufacturer supplying PC retailer.
parameter	supplier	reseller															
mutuality	medium	medium															
significance	medium	medium															
control	low	high															
informedness	medium	medium															

continued ...

Table 1 continued.

End supply	 <table border="1"> <thead> <tr> <th>parameter</th><th>supplier</th><th>customer</th></tr> </thead> <tbody> <tr> <td>mutuality</td><td>high</td><td>low</td></tr> <tr> <td>significance</td><td>low</td><td>medium</td></tr> <tr> <td>control</td><td>low</td><td>high</td></tr> <tr> <td>informedness</td><td>low</td><td>low</td></tr> </tbody> </table>	parameter	supplier	customer	mutuality	high	low	significance	low	medium	control	low	high	informedness	low	low	The essential factor of end supply is that the supplier cannot make many assumptions about the customer. Different customers will have different amounts of knowledge of the product they are procuring; in many cases non-expert customers are heavily reliant upon the advice given by the supplier. End supply implies acting as a single point of contact for the purchase of a range of goods and services, therefore an end supply has responsibilities in all four areas. However, end suppliers tend to advertise themselves as much as the products/services they sell. Example PC retailers supplying end-customers.
parameter	supplier	customer															
mutuality	high	low															
significance	low	medium															
control	low	high															
informedness	low	low															
Trading	 <table border="1"> <thead> <tr> <th>parameter</th><th>component supplier</th><th>manufacturer/ service creator</th></tr> </thead> <tbody> <tr> <td>mutuality</td><td>medium</td><td>low</td></tr> <tr> <td>significance</td><td>medium</td><td>medium</td></tr> <tr> <td>control</td><td>low</td><td>low</td></tr> <tr> <td>informedness</td><td>low</td><td>low</td></tr> </tbody> </table>	parameter	component supplier	manufacturer/ service creator	mutuality	medium	low	significance	medium	medium	control	low	low	informedness	low	low	Trading is a form of co-operation between competitors in a market. As the two parties involved are peers, they have a great deal in common, the need for support is minimal, and this makes this form of supply very lightweight. Trading relationships are often bi-directional, an enterprise will buy and sell via this interface. These kinds of relationships can be either formalised or <i>ad hoc</i>. Example: Interconnect agreements between network operators.
parameter	component supplier	manufacturer/ service creator															
mutuality	medium	low															
significance	medium	medium															
control	low	low															
informedness	low	low															
Commission	 <table border="1"> <thead> <tr> <th>parameter</th><th>supplier</th><th>agent/broker</th></tr> </thead> <tbody> <tr> <td>mutuality</td><td>medium</td><td>medium</td></tr> <tr> <td>significance</td><td>medium</td><td>medium</td></tr> <tr> <td>control</td><td>low</td><td>high</td></tr> <tr> <td>informedness</td><td>medium</td><td>medium</td></tr> </tbody> </table>	parameter	supplier	agent/broker	mutuality	medium	medium	significance	medium	medium	control	low	high	informedness	medium	medium	Commission supply is when a supplier relies on a third party (agent or broker) for some aspects of the supply conversation. The supplier and third party are only paid in the event of the end-customer making a purchase, therefore it is in both parties' best interests that a sale is made. The diagrams refer to the supply of the product being brokered rather than the supply of the brokerage service. The amount that the third party is involved within the supply conversation is used to differentiate brokers and agents. Brokers facilitate the sale in that they advise the customer and put the customer in contact with a suitable supplier, they have no post sales responsibilities. Agents, however, take part in the transaction and do have post-sales responsibilities, e.g. travel agents. The varied nature of brokerage means that there is insufficient space within this paper to cover this form of supply in detail. Therefore, for the remainder of the paper commission supply will be considered a specialised form of resell supply.
parameter	supplier	agent/broker															
mutuality	medium	medium															
significance	medium	medium															
control	low	high															
informedness	medium	medium															

- advertising (A),
- sales (S),
- revenue collection (R),
- post-sales support (P),

are used as examples. The arrows in the figures in Table 1 indicate a support responsibility in relation to what is supplied to the customer, e.g. a retailer (end supply) may only have a minor role (dashed line) in advertising the products they sell, the main responsibility (solid line) for this advertising being owned by the manufacturer. The arrow representing the support responsibility for collecting revenue points to the payer, and therefore revenue flow is in the opposite direction to the arrow. All enterprises have these responsibilities in relation to the supply of their customers. However, some supply stamps imply additional support responsibilities in relation to their customer's customer.

The output stamps are summarised in Table 1. What they show is that the mode of supply affects not only the number and type of customers an enterprise will have, but also the support responsibilities an enterprise has in relation to those supply relationships. For example, just because an enterprise does not supply the end customer, it still may need to advertise to the end customer. These support responsibilities need capabilities and systems to fulfil them.

4. Supply chain issues

With these building blocks, the establishment of supply chains can begin and the interactions between the links within them analysed. More importantly, future supply chains can begin to be predicted and the resultant impact of these supply chains on support systems understood. When considering the different supply chains that may exist for the delivery of a range of services, the full range of supply stamps to which an enterprise may need to conform can be seen.

Taking the example of the mobile telephony provider (service creator) that operates a network capable of allowing people on the move to be contacted, a number of possible supply types exist. This service creator could supply its service to the following enterprises in the following ways.

- Fleet management service — component supply

This enterprise offers a fleet management service that allows a fleet manager to monitor the location of a fleet of vehicles and request details such as current load, average speed, and current route. This information is transmitted from the vehicles to the control centre by sending data over the mobile telephony service. In this

case, the mobile telephony service is an imbedded part of the final product — the end customer may not even be aware of how the data is transmitted.

- Mobile Internet supplier — aggregator supply

This enterprise offers a complete solution for accessing the Internet while 'on the move'. They supply the required equipment, mobile telephony services and ISP functionality. The mobile telephony service is therefore an aggregated element. The customer could choose to change the telephony provider while retaining the ISP.

- Mobile telephony service — resell supply

The mobile telephony service is purchased wholesale and sold retail.

- Mobile telephony customer — end supply

The mobile telephony service provider also sells direct to the end customer.

The above example shows how network services may need to be provided in various ways to meet the differing needs of the service provider's customers.

Figure 1 describes a possible breakdown of the provision of Internet services to the market. It involves a number of roles and players and a variety of supply stamps.

GameNet are a specialist ISP for the games player. They offer a package that includes Internet access and subscription to a range of games, some of them bought in (aggregation) from GamesCo. They also resell voice over IP (VoIP) conference services, sourced from a different (resell) supplier.

In Fig 1, customer A is using dial-up connection to GameNet to gain access to the VoIP conferencing services that are used with customer B. GameNet use a Lo-call or Freephone service that is supported by Net2, modem banks supplied by PortsRUs and is connected to the Internet backbone through BackboneCo.

This example only shows one of the many types of supply chain that could exist. Other topologies may develop where retailing or wholesaling resellers or agents and brokers (service creators) act as intermediaries within one part of the competitive supply chain, e.g. network connectors to network service providers. There is perhaps a 'waterfall' style development of the supply stamps (see Fig 2). Component supply can be turned into all other types of supply, whereas end supply cannot be changed. Understanding the nature of a competitor or a market can potentially allow inefficiencies in that market to be identified, e.g. the potential for a resell supply instead of component supply between the network connectors (NetRUs and Net2) and the network service providers.

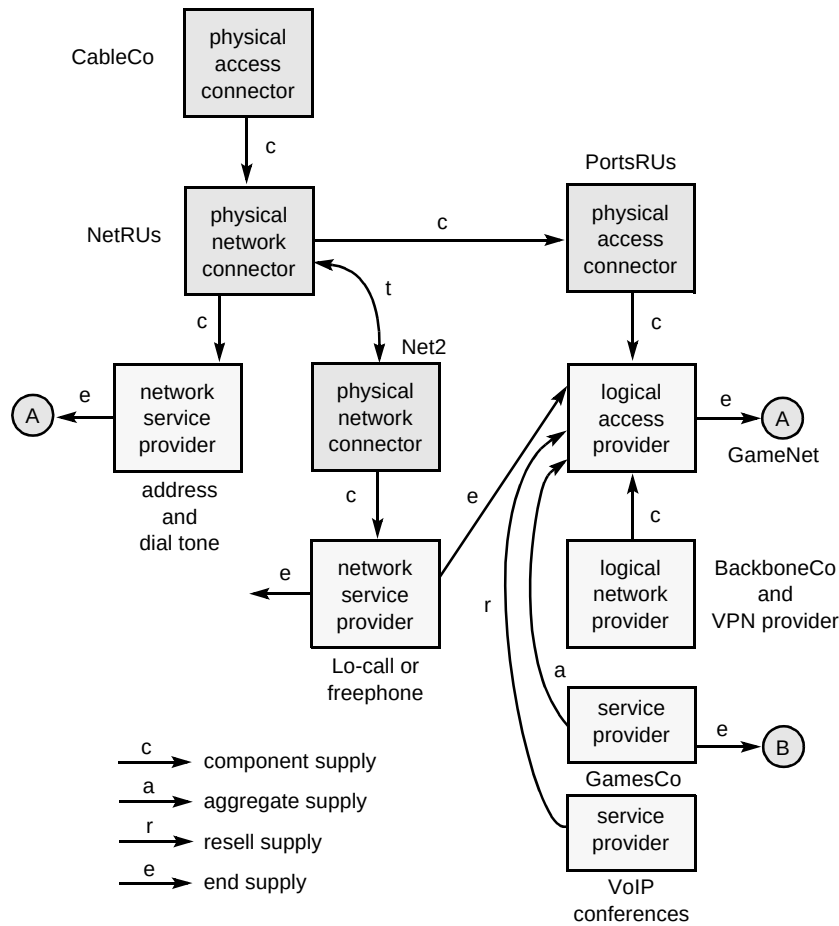


Fig 1 Example supply chain.

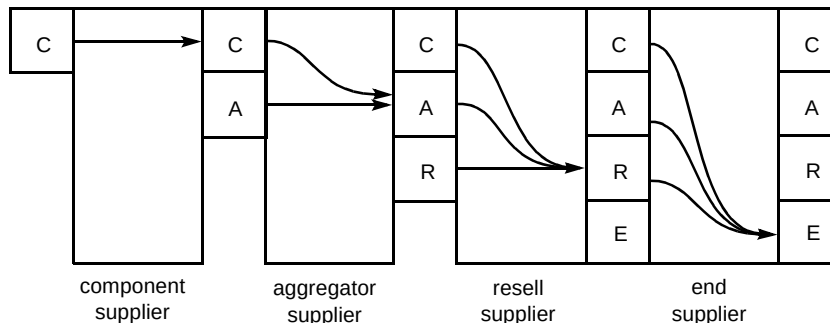


Fig 2 Supply chain 'waterfall'.

What this waterfall model implies is that intermediaries within the supply chain need to ensure that their mode of procurement and supply are appropriate for the supply chain to work efficiently. A reseller who procures via an end-supply stamp will be at an efficiency disadvantage compared with a reseller that procures via a resell stamp.

5. Operational support issues

The business stamps help understand the responsibilities that an enterprise has to undertake in its dealings with another enterprise, and the requirements of the enterprise's support systems. Each of the transformation stamps requires key operational support, for example:

- manufacturer — production control, procurement,
- service creator — service resource management,
- reseller — stock control, procurement.

However, the supply stamps can also be used to identify the relative importance of various operational support activities that either directly or indirectly support the customer relationship. The categories of operational support activities used here (see Table 2) were developed as part of the Eurescom project P846 [6].

Table 2 Operational support activities.

Sales and order processing (S&OP)	This includes pre-sales advice and advertising, the sale itself and the resultant processing.
Service instance provision (SIP)	Configuration of the service resource to meet orders.
Quality assurance (QA)	Includes post-sales support, such as fault handling, customer enquiries.
Service resource management (SRM)	Ensuring efficient use of the resource used to provide the service.
Account collation (AC)	Determining how much the customer owes or is owed.
Chargeable event collection (CEC)	Measuring the use of the resource (service) for the purposes of charging.

Referring back to the supply stamps, the impact that particular supply stamps have on these activities can be characterised. Table 3 shows the relative importance, from the customers' perspective, of the operational support activities for the different stamps. The relative importance of these activities gives a 'shape' to the required operational support from the customers' perspective. This 'shape' adds to the descriptions in Table 1 to fully characterise the stamp.

As can be seen from Table 3, the 'shapes' of the operational activities for each supply stamp are different. Figure 3 presents these different 'shapes' for comparison. Whereas end supply involves significant operational support, trading does not. Component supply is dominated by the need to maintain quality (QA, SRM). Resell supply places more importance on revenue collection (AC, CEC) and selling and provisioning (S&OP, SIP). Different industries and different players may produce different shapes depending on a number of factors, such as the nature of the product/service and the market in which it is offered. This paper presents 'stereotypical' shapes that were derived from the conversational metrics presented in section 3, for instance the high mutuality associated with component supply creates an emphasis on quality assurance. There are a number of alternative methods that could be used to determine the shapes for particular supply relationships, depending on whether the stamps were to be used predictively or for post-implementation analysis. The purpose of this paper is to show the relationship between the nature of supply and the nature of operational support, and therefore a discussion of these approaches is outside its scope. In summary:

- different supply stamps require different ranges of operational support,
- end supply is the most operational-support-intensive supply stamp to adopt,
- whereas component supply and resale supply have similar overall operational support requirements, the emphases are very different,

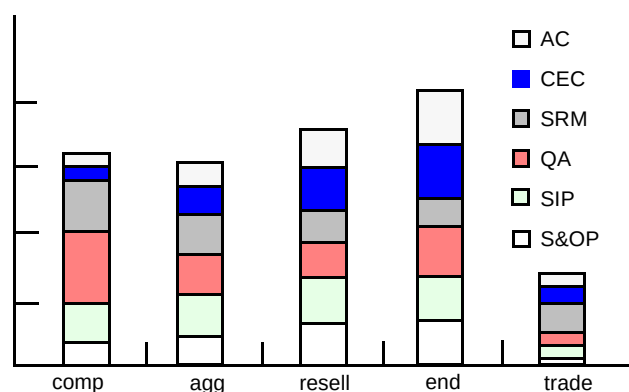


Fig 3 Operational support shapes for different supply stamps.

- some activities are of reasonably uniform importance for all supply types, such as service resource management (SRM) — others change significantly depending on the supply type, e.g. account collation (AC).

6. Discussion

The use of enterprise supply stamps helps link the design and scope of the business or service offering to that of the operational support systems. The illustrative example described above shows how one can influence the other. By identifying the supply stamps to which the enterprise will conform, the operational support systems can be designed to meet the business needs of both the customer and supplying enterprise. It also shows the implications to the operational support design of offering an existing service via a different supply stamp; using the same operational support systems for different supply stamps may prove to be a sub-optimal solution.

The new competitive environment within the telecommunications industry means that telcos are also considering buying-in services from other companies. Therefore, the relevant form of procurement is also required. If one company wishes to aggregate the service offering of another, it is important that the commercial as well as technical relationships are fit for purpose. Just as it is inappropriate for Ford to buy their components over the counter from a DIY superstore, it is also inappropriate for a network operator to buy-in value-add services with end-user supply.

There is an implied order to the different supply stamps (component, aggregation, resell and end supply). Each step in the line is a step closer to the end customer. In the general case it is inefficient for an enterprise to procure at one level of this chain and supply at a higher level. Component suppliers should only procure via the component supply stamp, whereas resellers can procure via any stamp other than end supply.

Table 3 Operational support shapes for different supply stamps.

Stamp	Notes	Shape of operational support activities
Component supply	The high mutuality and significance of the supply conversation implies that ensuring the supplied service operates to specification is vital (QA). The high informedness and significance implies richness to the sales activity that is not suitable to be delegated entirely to an operational support system. However, the control required by the component customer may mean that they wish to be able to perform service instance provision themselves. The limited number of these conversations implies that the requirement for account collation and chargeable event collection is minimal.	
Aggregator supply	As stated below the mutuality within aggregation is less than that associated with component supply, and this is reflected in the lower importance of the QA activity. Similarly, the coupling between the aggregated element and the end product implies that the supplied element will be more 'off-the-shelf', (lower informedness). Therefore, the S&OP activity may become more important. The nature of the aggregation may be such that the supplier of the aggregated product may charge the customer for usage direct, thereby increasing the importance of the CEC and AC activities.	
Resell supply	QA activities are less important for this form of supply, this is because the reseller will take on some of the QA responsibilities and because the mutuality is so much less. The nature of the relationship with resellers will be much more distant (less informedness, mutuality), and transactions may be via a number of repeated orders rather than deliveries made against a single contract. As the supplier will have less control, sales and order processing activities need to be more responsive and efficient.	
End supply	End supply can be characterised as a large number of low significance, low informedness interactions. This makes the revenue-gaining activities (S&OP, SIP, CEC, AC) very important. The end user may also be a non-expert user relying on the supplier (high mutuality); this increases the importance of those activities that assist the customer in use of the service (QA).	
Trading	As stated previously, the trading interface is a lightweight interface. As the parties are peers within the same industry there is little emphasis on the sales activities. There is little need to support the trading partner (QA) as they are suppliers of what they are consuming (low informedness). For longer term trading relationships, CEC and AC become more important as they need to deal with the flow of revenues both ways across the interface.	

Many of these issues are clear within a market environment but may be lost in 'internal' supply relationships where one part of a company may be supplying a different part of the same company. In Fig 4(a), the vertically integrated approach, where internal supply is not considered in the design (and the operational support is built equivalently), is compared to a modular (service-type) design that considers what type of supply may exist for each component 'business'. The example relates to that in Fig 1. The transition from the vertical to the modular approach to support the ability to 'unbundle' or sell into new markets is fraught with problems. The modular design allows specific operational support implementations tailored to the supply stamp, as in Fig 4(b), to be constructed (this does not imply that separate systems are constructed — reuse of common facilities is not precluded). The same service can then be offered efficiently to the different markets.

So far two conclusions have been reached.

- different supply stamps require different operational support capabilities,
- a service provider may need to conform to multiple supply stamps.

A single service may therefore need a number of different support systems, or tailored versions of a single system. This is particularly relevant to network services which, as the market matures, are being treated more as commodities. The quality of the support systems that deliver these services to the customer is becoming the only way to differentiate one network service provider from another. Understanding the needs of a particular customer type, and meeting those needs, is key to gaining competitive advantage.

This is indicative of a changing role for operational support and its progression to a 'front of house' position.

Rather than considering operational support as something the customer never sees, it can be viewed as part of the service that is offered to the customer. This is exemplified by 'services' such as View My Bill™ [7], and the CarrierNet [8] system which is an operational support system tailored to deliver BT's services to customers who are not just end users, i.e. other licensed operators (OLOs).

7. Conclusions

This paper has introduced and characterised a new pattern set to describe the nature of a business and how it relates to its market. It has also shown how these patterns affect the operational support activities that will be needed. This opens opportunities for the use of these business patterns and supply stamps:

- new opportunities can be identified by knowing the nature of the supply between the players within the market and supply chain (section 4),
- the operational support requirements of the company can be understood from the types of supply the business operates (section 5) — for new markets, an assessment of the cost of adopting a new supply stamp to approach a new type of market can be made.

Future companies will only be successful if they organise their operational support against the types of supply that they adopt for their market offering.

The successful company will understand its position(s) in the supply chain and will devise new products that conform to their supply type(s). Such patterns will be core to a flexible strategy within high performing, adaptable businesses.

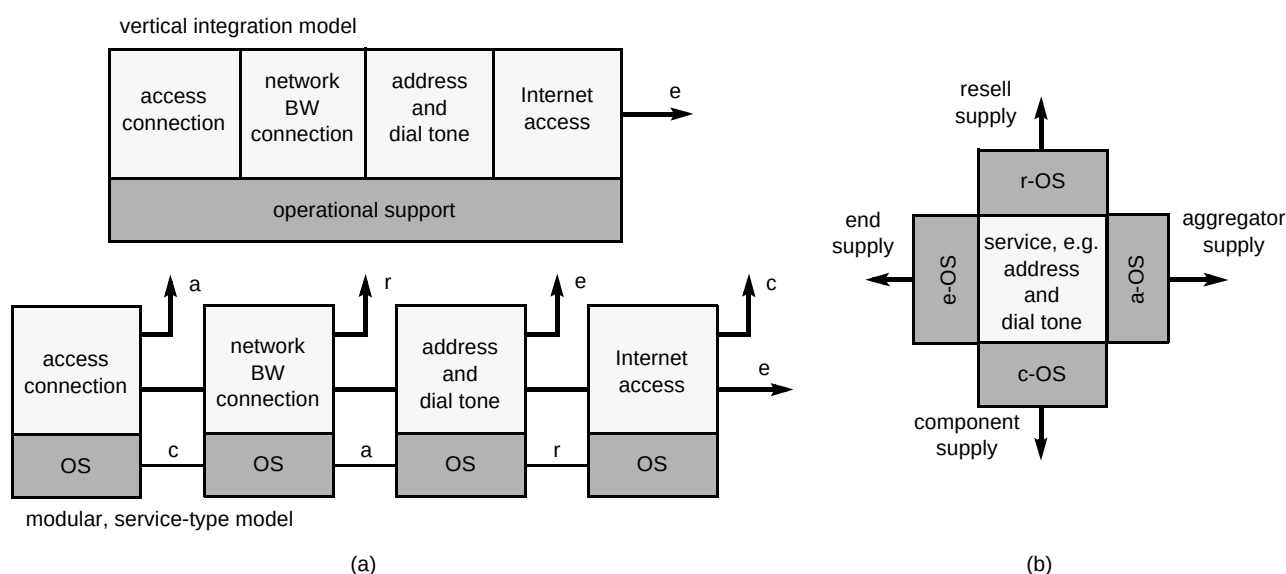


Fig 4 Vertical and modular designs.

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